

HULTSFRED, SwedenWMO: **025650**Lat: **57.526N**Lon: **15.823E**Elev: **366**StdP: **14.50**Time Zone: **1.00 (EUC)**Period: **08-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.4	9.3	-1.3	5.2	2.7	6.6	7.8	12.9	22.2	41.7	19.7	40.4	1.8	280	0.585

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.3	81.8	64.2	78.3	62.9	75.0	61.2	67.1	76.6	65.3	74.1	63.5	71.5	7.8	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.1	90.8	69.8	62.1	84.5	68.0	60.2	79.0	66.5	31.7	77.2	30.4	73.9	29.0	71.5	72.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.4	16.1	14.1	DB	-3.6	86.6	9.6	4.2	-10.4	89.6	-16.0	92.1	-21.4	94.4	-28.3	97.5	(4)
(5)			WB	-1.9	69.7	8.5	1.9	-8.0	71.1	-13.0	72.2	-17.8	73.2	-24.0	74.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	44.7	27.8	29.9	35.3	42.7	51.1	57.7	62.5	60.0	53.7	44.7	38.3	31.8	(6)	
	DBStd	13.56	8.67	8.31	7.32	6.24	6.79	5.44	5.01	4.57	5.35	6.21	6.52	9.62	(7)	
	HDD50	3138	688	564	455	230	68	5	0	0	25	185	352	566	(8)	
	HDD65	7466	1153	984	919	670	432	229	112	166	341	629	801	1030	(9)	
	CDD50	1202	0	0	0	10	102	235	387	311	135	21	1	0	(10)	
	CDD65	55	0	0	0	0	1	8	34	11	1	0	0	0	(11)	
	CDH74	923	0	0	0	0	58	207	483	161	14	0	0	0	(12)	
	CDH80	183	0	0	0	0	3	35	108	36	1	0	0	0	(13)	
(14) Wind	WSAvg	6.2	6.3	6.5	7.0	6.4	6.0	6.3	5.7	5.5	6.0	6.1	6.4	6.6	(14)	
(15) Precipitation	PrecAvg	24.3	1.5	1.4	1.1	1.3	1.8	2.6	3.5	2.6	2.4	2.1	2.2	1.9	(15)	
	PrecMax	29.8	2.7	2.7	1.9	3.0	3.4	5.4	7.8	5.1	6.1	4.3	5.5	3.4	(16)	
	PrecMin	18.9	0.2	0.3	0.1	0.0	0.3	0.2	1.0	0.3	0.4	0.7	0.4	0.4	(17)	
	PrecStd	2.9	0.7	0.6	0.5	0.8	0.8	1.2	1.6	1.2	1.6	1.1	1.1	0.8	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.7	50.0	60.6	70.5	79.2	84.5	87.1	84.3	76.2	65.1	55.0	51.3	(19)	
		MCWB	44.6	41.8	48.6	53.5	61.6	64.1	66.4	65.7	62.5	56.7	52.2	48.4	(20)	
	2%	DB	43.2	45.8	55.0	65.2	75.2	79.5	82.9	78.1	70.3	59.6	51.6	48.6	(21)	
		MCWB	41.1	41.3	45.0	51.0	58.6	61.8	64.8	64.7	60.2	54.9	49.3	46.2	(22)	
	5%	DB	41.0	42.8	50.1	60.6	71.4	75.6	79.4	74.2	66.9	56.9	49.3	45.6	(23)	
		MCWB	39.0	39.8	42.9	48.8	56.8	60.2	64.0	62.1	58.0	52.9	47.2	43.2	(24)	
	10%	DB	38.7	40.4	46.3	56.4	66.7	71.7	75.7	71.4	63.7	54.6	47.1	42.8	(25)	
		MCWB	37.0	38.0	41.0	46.8	54.8	58.1	62.3	60.6	56.5	51.0	45.4	40.8	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.7	44.5	49.6	55.5	63.4	67.8	69.8	69.2	65.0	58.4	52.6	48.7	(27)	
		MCDB	46.4	47.6	57.9	66.4	74.8	77.9	79.2	77.5	72.4	62.4	54.1	50.8	(28)	
	2%	WB	41.6	41.9	46.7	52.7	61.0	64.7	67.5	66.3	61.7	55.6	49.4	46.5	(29)	
		MCDB	43.0	44.7	54.7	62.7	71.3	75.0	77.6	75.2	67.5	58.8	51.1	48.5	(30)	
	5%	WB	39.3	40.0	43.9	50.1	58.5	61.8	65.8	64.3	59.7	53.6	47.5	43.5	(31)	
		MCDB	40.8	42.6	49.0	58.8	67.3	72.4	75.3	71.1	64.7	56.6	49.0	45.2	(32)	
	10%	WB	37.0	38.1	41.7	47.9	55.7	59.4	64.2	62.5	57.9	51.6	45.7	41.1	(33)	
		MCDB	38.5	40.5	46.0	55.8	64.7	69.1	72.5	69.1	62.1	54.2	47.0	42.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.9	11.3	16.2	21.7	23.2	22.2	21.3	20.8	18.0	13.8	9.1	9.6	(35)	
		MCDBR	10.1	14.1	22.3	31.0	31.4	30.6	28.3	25.8	23.2	16.5	11.3	10.2	(36)	
	5% WB	MCWBR	9.0	10.4	13.9	17.8	16.1	14.5	12.4	13.4	13.6	11.8	10.0	9.4	(37)	
		MCWBR	9.7	12.6	20.1	26.5	25.8	26.6	23.0	21.7	19.6	15.5	10.2	10.1	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.289	0.321	0.338	0.375	0.373	0.362	0.389	0.382	0.351	0.341	0.316	0.277	(40)	
		taud	2.283	2.324	2.360	2.327	2.394	2.448	2.392	2.407	2.482	2.429	2.328	2.227	(41)	
	Ebn at Noon		183	229	258	265	274	278	267	260	253	220	170	149	(42)	
		Edh at Noon	15	23	28	34	34	33	34	31	25	20	15	12	(43)	
(44) All-Sky Solar Radiation	RadAvg		116	309	783	1280	1669	1808	1636	1322	899	409	152	83	(44)	
	RadStd		20	61	83	163	150	91	191	128	117	71	28	10	(45)	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (48 neighbors)		+0.76	N/A	N/A	N/A	N/A	N/A	-225	-292	N/A	N/A

Nomenclature: See separate page